

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	BANDI MOHITH MANOJ	Reg. No.:	192472018
Section / Batch:	BE-CSE-AI	Date:	17-08-2026

Code of Conduct

I BANDI MOHITH MANOJ - 192472018(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

- 1) Recursively production rules allow hierarchical sentence generation - (B) ✓
- 2) (a) ✓
- 3) (b) ✓
- 4) (b) sub categorization frame ✓
- 5) (b) Left recursion causing ✓
- 6) (b) uses predictor, scanner and completer operations to process partial input efficiently. ✓
- 7) (a) Left factoring X
- 8) (a) chooses the parse tree with the max true probability. ✓
- 9) (b) Early parses + PCFG ✓
- 10) (A) - CFG + Feature Structures + sub categorization + PCFG. ✓
- 11) (b) Add the recursive productions $NP \rightarrow NP PP$ ✓
- 12) (A) The list of authorized employees were updated. X
- 13) (C) The verb violates the sub categorization frame. X
- 14) (A) Ambiguous lexical entries. X
- 15) (D) Dynamic chart construction Combining predictor, scanner and completer. ✓
- 16) A - number feature conflict. X
- 17) A - Tree, because it has higher probability. ✓

$$\frac{36}{50}$$

18 (c) - well factored CFG with minimized ambiguity ✓

19 (D) - Early parsing handles ambiguity efficiently, X
while PCFG ranks

20 (C) - CFG parsing - subcategorization checking ✓

21 (C) - Introduce $N \rightarrow \text{"Living room"} N$ ✓

22 (b) - object noun phrase. ✓

23 (b) - Feature unification. ✓

24 (A) - shift reduce conflict (X) ✓

25 (C) - Completer. ✓

26 (B) - Maximum likelihood estimation. X

27 (B) - Tree B because it has fewer productions. ✓

28 (A) - CFG. X

29 (B) - Recursive Top-Down Parser ✓

30 (B) - Highest PCFG rule probability. X

31 D - S - NP VP - NVP $\rightarrow$ robot delivers package. X

32 (D) - Engineers Develops software ✓

33 (C) - she gave the customer a refund. X

34 (C) - Build parse trees. ✓

35 (A) - predictor ✓

36 (B) - Features are failed. X

37 (B) - The shortest parse tree ✓

38 (A) - Shift reduce parsing ✓

39 (A) - parse A. has highest probability. ✓

40 (B) - Reject the parse. X

41 (C) - $S \rightarrow NP VP$ ✓

43 (C) - predictor ✓

42 (B) - VP. ✓

44 (A) Sentence accepted because person matches. X

45. (A) ✓

46. (A) CFG supports recursive noun phrase generating using ✓

47. (C) structural (syntactic ambiguity) ✓

48. (B) Left recursion ✓

49. (A) The verb gave required additional complements. ✓

50. (C) V. NP PP ✓

Annexure A – MCQ Test

<https://forms.gle/WSv28zfaU5B2aWcy6>

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____